

Ferroptosis Inducing Drugs: A New Way to Get a Leg Up on Chemo and Radiotherapy Resistance

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Abstract

This article presents information from studies, trials, and literature about ferroptosis, ferroptotic inducing drugs, chemotherapy resistance, and radiotherapy resistance in cancer. It goes into detail about the ferroptosis pathway and how it emerged as a new way to override chemo and radiotherapy resistance in cancer. It also discusses different drugs and compounds that could induce ferroptosis. Some of the drugs and compounds explored in this article have been created with ferroptosis in mind, others were repurposed from different drug classes, or they could be drugs that have an enhanced effect when used in tandem with other drugs as opposed to being used alone.

Introduction

What is Ferroptosis

Ferroptosis is a non-apoptotic form of regulated cell death that requires intercellular iron, especially Fe^{+2} . It happens when ROS causes the accumulation of toxic lipid hydroperoxides on polyunsaturated fatty acids (PUFAs), especially phosphatidylethanolamine. When these toxic lipids accumulate, the cell membrane loses its integrity.

ROS can be produced enzymatically or non-enzymatically. Non enzymatic lipid oxidation can involve iron dependent Fenton chemistry, while enzymes such as lipoxygenases can also cause lipid peroxidation.

Ferroptosis can be prevented by iron chelators such as deferoxamine or by lipophilic antioxidants. It can also be induced by several ferroptosis inducers, including Erastin, RSL3, sulfasalazine, sorafenib, artesunate, FIN56 and FINO2.

Main Ferroptosis Pathways

One of the most important pathways is system XC-, which is made mainly of SLC7A11 and SLC3A2. This system brings cystine into the cell.

Cystine is needed to make glutathione (GSH). GSH is an important antioxidant and is also required for GPX4 activity. GPX4 protect the cell by converting toxic lipid hydroperoxides into less harmful lipid alcohols. Therefore:

System XC- inhibition decreases cystine which decreases GSH, and GSH decreases GPX4 activity which then lead to increase lipid peroxides and then ferroptosis.

Iron is also very important. Transferrin helps bring iron into the cell, while ferritin stores iron and prevents it from being available in its active form. NCOA4 can breakdown ferritin through a process called ferritinophagy, releasing Fe²⁺ and increasing ferroptosis sensitivity.

Other important ferroptosis related molecules include 15 lipoxygenase, FSP1, CoQ10, GCH1, ACSL4, and LPCAT3. FSP1 and CoQ10 act as antioxidant protection against lipid peroxidation, while ACSL4 and LPCAT3 help provide the PUFAs containing membrane lipids needed for ferroptosis.

NRF2 is another major protective factor because it increases the expression of antioxidant and anti ferroptotic molecules such as SLC7A11, GPX4 and ferritin.

Ferroptosis and Cancer

Cancer cells can be particularly sensitive to ferroptosis. And usually grow rapidly and have high energy requirements, which can result in higher ROS levels than normal cells. Because of this, cancer cells become more dependent on the antioxidant system to survive.

SLC7A11 is also abnormally expressed in many cancers. Blocking it may therefore be a way to target cancer cells while having less effect on normal cells. They also tend to depend more on iron than normal cells. Some cancer associated mutations including RAS and EGFR can increase ferroptosis sensitivity and are associated with increased dependence on NRF2 activity and antioxidant dependence.

Overall, the paper suggests that cancer cells depend heavily on mechanisms that protect them from ferroptosis, making these mechanisms potential therapeutic targets.(1)

Chemotherapy resistance and cancer

One paper demonstrated how resistance to chemotherapy (in this case, carboplatin) in retinoblastoma is acquired and its effect on ferroptosis. It was hypothesized that SALL1 activates the transcription of Microsomal glutathione S-transferase 1 (MGST1), which is an enzyme that catalyzes the conjugation of glutathione with electrophilic substrates and possesses glutathione peroxidase activity. Studies have demonstrated that MGST1 associated with chemotherapy resistance and can impede the ferroptosis process by inhibiting lipid peroxidation. This suggests that enhanced antioxidant protection against lipid peroxidation may be a contributing factor in chemotherapy-resistant cell survival. Consequently, it could be possible to bypass the resistance by pharmacologically inducing ferroptosis through increasing lipid peroxidation and exploiting other vulnerabilities that exist in the resistant cells.(2)

Ferroptotic inducing drugs

Several studies explored multiple different compounds which could act as an anti-cancer through ferroptotic induction, and one of them is a mild photothermal immunotherapy nanoplatfrom (termed Fe-Mn@GOx NPs), designed to induce non-apoptotic ferroptosis, augment cGAS STING pathway activation and enhance innate anti-tumor immunity.(3)

Fe-Mn@GOx NPs can act as both a ferroptosis trigger and a cGAS-STING pathway agonist. It possesses a Fe-Mn bimetallic catalytic center and exhibit excellent catalytic activity, enabling in situ conversion of endogenous H₂O₂ to cytotoxic •OH, thereby degrading HSPs to potentiate transition metal-mediated mild photothermal therapy. It is also capable of depleting intracellular glutathione (GSH), the pathway is bidirectionally activated. This strategy mediated by Fe-Mn@GOx NPs awakens innate immunity through the activation of the cGAS-STING signaling pathway, establishing a novel perspective for immunotherapy refractory of triple-negative breast cancer.

Researchers found that the tumor-derived endogenous DNA in the debris of dead cells caused by Fe-Mn@GOx NPs activates the cGAS-STING pathway in neighboring immune cells, initiating downstream signaling. Moreover, there was found that no obvious tissue damage or inflammation in the main organs (heart, liver, spleen, lung and kidney), confirming that Fe-Mn@GOx NPs have good biosafety and biocompatibility.(3)

In a different study, it focused on oxaliplatin (OXA) resistance in pancreatic cancer and how to overcome this resistance using ferroptosis induction such as artesunate (ART). The researches established 42 patient derived pancreatic cancer organoids (PDPCOs) by using ultrasound guided fine needle biopsy. After that, they performed drug response experiments and based on this they classified PDPCOs into sensitive and resistance groups, then they treated 3 cell lines with ART alone or oxaliplatin alone and in combination to see if it could improve the response. They assessed cell viability, lipid peroxidation, intracellular iron levels and ROS. Moreover, the researchers studied SLC7A11 by increasing and decreasing its expression and used SLC7A11 inhibitors such as erastin or sulfasalazine. The treatments were also assessed in a mouse xenograft.

The resulting observations were that the combination therapy of ART and OXA showed a strong synergetic effect. Furthermore, it was demonstrated in the study that silencing SLC7A11 made resistant cells more sensitive to the treatment. Therefore, erastin, a selective SLC7A11 inhibitor, was used in a triple combination therapy with OXA and ART to inhibit SLC7A11 and overcome resistance.(4)

Moreover, one paper mentioned MLN0128, an ATP-competitive mTOR kinase inhibitor, and PD901, an ERK inhibitor. It explored the therapeutic effect of combining MLN0128 and PD901 on hepatocellular carcinoma, where the results demonstrated that the combination of both drugs inhibited tumor cell proliferation by suppressing the mTOR signaling pathway. Additionally, the combination displayed a synergistic effect on ferroptosis compared to monotherapy, and this provides a new potential way for overcoming acquired mTOR inhibitor resistance in HCC and to improve 2nd line treatment outcomes.(5)

On another note, a different study investigated the repurposing of the H1-antihistamine Loratadine for targeting MDR1-overexpressing cancer cells (KB-V-1 line) by inducing ferroptosis. In the study, Loratadine (LOR) selectively inhibited the growth of drug-resistant KB-V-1 cells compared to chemosensitive KB-3-1 cells. The experiment displayed that LOR exerted its action via a double mechanism: ER stress activation, and downregulation of MDR1. In the first mechanism, it activated the endoplasmic reticulum stress axis and upregulated targets of ATF4, such as CHAC1 DDIT3, and DDIT4 to amplify cellular redox collapse. On the other hand, Loratadine downregulated the protein content in MDR1/P-glycoprotein, a membrane transport protein that pumped out drugs and shrank drug efflux capability from tumor cells.(6)

While on the topic of repurposing drugs, one study used Auranofin, an anti-rheumatoid arthritis drug, as a ferroptosis inducer on SCL/NEL (stem cell-like/ neuroendocrine-like) cells in vitro and in vivo, in an effort to target and inhibit AP-1 transcriptional activity in order to decrease GPX4 expression thus inducing ferroptosis in enzalutamide-resistant prostate cancer.(7)

In other regards, scientists have derived compounds from marine life capable of inducing ferroptosis. *Holothuria scabra* is a sea cucumber where its derived compounds have anti-proliferative and anti-metastatic potential in osteosarcoma. It was examined for treatment for osteosarcoma, and it was observed that there was a notable decrease in GPX4, NRF2, and GSH levels. As mentioned previously, these changes are the hallmark of ferroptotic cell death. However, simultaneous NRF2 suppression is required as NRF2 is a pivotal part in the regulation of antioxidant defenses and ferroptosis resistance. (8)

Radiation and Ferroptosis

Finally, the relationship between ferroptosis and treatment resistance may not be restricted to chemotherapy; radiation increases ROS, lipid peroxidation, and intracellular Fe²⁺, which can promote ferroptosis. It can also increase ACSL4, which helps produce PUFA containing lipids (1), therefore, exhibit increased vulnerability to ferroptotic death because these specific lipids are

highly susceptible to iron-dependent lipid peroxidation (9). P53 can reduce SLC7A11 after radiation, decreasing GSH and allowing peroxide to accumulate. However, tumor cells can resist this by increasing antioxidant pathways such as SLC7A11, GPX4, CD44v, and NRF2. These pathways reduce lipid peroxidation and help cancer cells survive radiation.

Ferroptosis Inducers as Radiosensitizers.

Ferroptosis inducers (FINs) can block these protective mechanisms and make tumors more sensitive to radiation.

Class1: Erastin, sulfasalazine, sorafenib inhibit SLC7A11

Class2: RSL3, ML162 inhibit GPX4

Class3: FIN56 affects GPX4 AND CoQ10.

Most studies showed that FINs combined with radiation increased lipid peroxidation and reduced tumor cell survival.

Important FINs

Erastin: inhibits SLC7A11 and increased radiation sensitivity in several cancer models. IKE, an Erastin analog, also showed promising results, but neither has been tested in human trials.

RSL3: directly inhibits GXP4 and increases radiation induced lipid peroxidation and cell death.

BSO: reduces GSH production and increases radiation sensitivity.

Nanoparticles: iron containing nano particles can induce ferroptosis and may improve radiosensitization while reducing toxicity.

FDA Approved Drugs

Sulfasalazine: Inhibits system Xc- introduces GSH. Combined with radiation, it increased cancer cell death in several models.

Sorafenib: can inhibit system Xc- and increase radiation sensitivity, but some clinical combinations cause significant toxicity.

Artesunate: increases ROS and lipid peroxidation and showed radiosensitizing effects in several cancer models.

Hemin: increases intracellular iron and reduces GPX4, promoting radiation induced ferroptosis.(1)

Conclusion

This article demonstrates the findings of the effects and uses of ferroptotic inducing drugs on chemo and radiotherapy resistant cancer. The results suggest that ferroptosis may be a way to bypass that resistance and stop tumor proliferation in its tracks. While this field of study has already showed promising results, more research, studies, and clinical trials are required to fill in the gaps to reach a more reliable way to eradicate cancer.

References

1. Carmicheal J, Fritson CM, Seas A, Vieira H, Batra SK. Overcoming Radiation Resistance: Ferroptosis Induction to Sensitize Solid Tumors to Radiation Therapy. *Cancer Med.* 2026 Jul 26;15(8):e72146. doi:[10.1002/cam4.72146](https://doi.org/10.1002/cam4.72146) PubMed PMID: 42504052; PubMed Central PMCID: PMC13402861.
2. Xiong Z, Zhang W, Yu X, Gu R, Zhang X. The role of SALL1-MGST1 axis-mediated ferroptosis inhibition in chemoresistance of retinoblastoma. *PLoS One.* 2026 Jul 6;21(7):e0353000. doi:[10.1371/journal.pone.0353000](https://doi.org/10.1371/journal.pone.0353000) PubMed PMID: 42406787; PubMed Central PMCID: PMC13336194.
3. Yuan H, Wang X, Bai YM, Ma J, Chen W, Chen J, et al. Dual-functional catalytic photothermal immunological agent for ferroptosis induction and cGAS-STING pathway activation: Enhancing tumor immunotherapy and generating robust immune memory. *Mater Today Bio.* 2026 May 20;38:103246. doi:[10.1016/j.mtbio.2026.103246](https://doi.org/10.1016/j.mtbio.2026.103246) PubMed PMID: 42293383; PubMed Central PMCID: PMC13253198.
4. Choi JH, Lee KM, Oh M, Noh J, Kim Y, Hwang E, et al. Ferroptosis-based therapeutic strategy targeting iron metabolism and SLC7A11 overcomes oxaliplatin resistance in patient-derived pancreatic cancer organoids. *Journal of Experimental & Clinical Cancer Research.* 2026 May 30;45(1):169. doi:[10.1186/s13046-026-03745-z](https://doi.org/10.1186/s13046-026-03745-z)
5. Cook N, Li X, Bowen NJ, Danaher A, Thomas M, Alhassan S, et al. Repurposing loratadine to induce ferroptosis and overcome multidrug resistance: preclinical evidence in KB-V-1 cells. *Front Oncol.* 16:1829405. doi:[10.3389/fonc.2026.1829405](https://doi.org/10.3389/fonc.2026.1829405) PubMed PMID: 42388634; PubMed Central PMCID: PMC13318782.
6. Zhou Y, He J, Zhang H, Wang Q, Yu Z, Zhang Y, et al. Ferroptosis vulnerability of enzalutamide resistant prostate cancer conferred by ACSL4 overexpression and GPX4 antagonism. *Cell Death Dis.* 2026 May 29;17(1):665. doi:[10.1038/s41419-026-08906-8](https://doi.org/10.1038/s41419-026-08906-8) PubMed PMID: 42215431; PubMed Central PMCID: PMC13421459.

7.

Li R, Zhou Y, Wang Z, Liu G, Fan D, Huang L, et al. Targeting the 4EBP1/HSP90 β /Nrf2 Axis Sensitizes β -catenin-mutant Hepatocellular Carcinoma to mTOR Inhibitors via Ferroptosis Induction. *J Clin Transl Hepatol*. 2026 Jun 28;14(6):649–64. doi:[10.14218/JCTH.2026.00072](https://doi.org/10.14218/JCTH.2026.00072) PubMed PMID: 42483739; PubMed Central PMCID: PMC13384959.

8.Sibarani JN, Khumaidi MA, Sakti YM, Permatasari HK, Hendrawan AF, Surya R, et al. Multi-Omics-Guided Discovery of *Holothuria scabra*-Derived Drug Candidates Targeting Ferroptosis and the Bone Tumor Microenvironment in Osteosarcoma. *Marine Drugs*. 2026 Jun 28;24(7):226. doi:[10.3390/md24070226](https://doi.org/10.3390/md24070226)

9.Soltani Tehrani S, Olson SI, Kundu K, Ferrandon S, Kalady MF. Targeting Ferroptosis to Overcome Radioresistance and Enhance Immunotherapy in Colorectal Cancer. *Cells*. 2026 May 28;15(11):993. doi:[10.3390/cells15110993](https://doi.org/10.3390/cells15110993) PubMed PMID: 42274587; PubMed Central PMCID: PMC13256493.